

CHAITANYA KAPOOR

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(Nationality: Citizen of the United States of America)

Research Interests

I develop computational models to understand how neural circuits implement perception and learning, combining tools from optimal transport, machine learning and representational geometry. My work bridges biological and artificial neural networks, with a focus on metrics and models that are both theoretically grounded and empirically testable.

Education

University of California, San Diego **Sep. 2025 – Present**
PhD Student (Cognitive Science) *La Jolla, CA*

University of California, San Diego **Aug. 2024 – Aug. 2025**
Post-Baccalaureate Student ([NeuroML Group](#)) *La Jolla, CA*
Advisor: [Meenakshi Khosla](#)

Massachusetts Institute of Technology **Jan. 2024 – July 2024**
Visiting Student ([Senseable Intelligence Group](#)) *Cambridge, MA*

Thesis: LUMIN: Light-sheet Microscopy Analysis Unified with Distributed and Domain-Randomized Generative Models

Advisor: [Satrajit Ghosh](#)

Birla Institute of Technology and Science, Pilani **Nov. 2020 – July 2024**
B.E. (Hons.) in Electrical and Electronics engineering *Pilani, India*

Selected Publications

Partial Soft-Matching For Neural Representational Comparison

C. Kapoor, A. H. Williams, M. Khosla

International Conference on Learning Representations (ICLR), 2026 - [paper](#), [code](#)

Bridging Critical Gaps in Convergent Learning: How Representational Alignment Evolves Across Layers, Training, and Distribution Shifts

C. Kapoor, S. Srivastava, M. Khosla

Neural Information Processing Systems (NeurIPS), 2025 - [paper](#), [code](#), [CCN Extended Abstract](#)

Multiplexed Expansion Revealing for Imaging Multiprotein Nanostructures in Healthy and Diseased Brain

J. Kang, M. Schroeder, Y. Lee, C. Kapoor, E. Yu, T. B. Tarr, K. Titterton, M. Zeng, D. Park, E. Niederst, D. Wei, E. S. Boyden

Nature Communications, 2024 - [paper](#), [code](#), [news story](#)

RnR-ExM: Robust Non-Rigid Registration Challenge for Expansion Microscopy Volumes

E. Besier, R. Zhang, Y. Bando, Y. Quémener, C. Kapoor, M. Alawi, M. Hoffman, A. Dalca, A.

Casamitjana, I. Arganda-Carreras, E. S. Boyden, H. Pfister, D. Wei

IEEE International Symposium on Biomedical Imaging (IEEE ISBI), 2023 - [code](#), [website](#)

Research Experience

Computational Neurobiology Laboratory **Jan. 2026 – Present**
Supervisor: Prof. Terrence J. Sejnowski *La Jolla, CA*

- Investigating traveling-wave-driven synaptic plasticity in large-scale sparse spiking networks. We propose a mechanistic link between cortical wave dynamics and Hebbian learning via temporally ordered spike sequences, STDP, and biophysical mechanisms including rebound inhibition and backpropagating action potentials.
- Developing a high-performance C++20 spiking network simulator handling millions of nodes with sparse connectivity, sub-millisecond timing precision, and structured binary I/O (custom spike-time formats at network scale).

NeuroML Lab

Supervisor: Prof. Meenakshi Khosla

Aug. 2024 – Present

La Jolla, CA

- Leading the development of an optimal-transport-based topographic similarity metric that jointly accounts for neural tuning similarity and cortical geometry over different metric spaces; validated on datasets spanning salt-and-pepper mouse V1 and columnar primate cortex.
- Led the development and design of new optimal transport-based representational similarity metrics allowing for matching and detection of single-neuron tuning outliers across biological and artificial neural networks. This work has been published at ICLR 2026.
- Led a large-scale study spanning CNNs, ViTs, and LLMs to characterize how internal representations converge across architectures; found that simple geometric invariances drive alignment, cross-network convergence emerges far earlier than previously thought, and alignment robustness degrades proportionally with distributional shift. This work is published at NeurIPS 2025.

Talmo Lab

Supervisor: Dr. Talmo Pereira

May 2023 – Aug. 2024

Salk Institute, La Jolla, CA

- Worked on the development of methods for pose estimation and tracking to quantify animal behavior through robust tracking of anatomical landmarks in 3D.
- Designed algorithms utilizing projective geometry to create inputs for a CNN that leverages geometric reasoning.

Senseable Intelligence Group

Supervisor: Prof. Satrajit Ghosh

Jan. 2024 – July 2024

Massachusetts Institute of Technology, MA

- Developed a distributed framework for large-scale ($\approx 1\text{TiB}$) image segmentation for light-sheet microscopy images.
- Designed a domain-randomized generative model, using spherical harmonics to synthesize cortical sections of *ex-vivo* human brains to enable zero-shot segmentation.
- Worked on designing few-shot, *spatially* guided semantic segmentation algorithms for MRI segmentation.
- Proposed the use of a topologically aware Graph Neural Network which is capable of encoding spatial and anatomical constraints.

Synthetic Neurobiology Group

Supervisors: Prof. Edward S. Boyden, Prof. Donglai Wei

Feb. 2022 – Jan. 2024

Massachusetts Institute of Technology, MA

- Worked on the development of a *generalist* 3D segmentation model for Expansion Microscopy (ExM) volumes for various animal species.
- Proposed a human-in-the-loop feedback learning mechanism, built on top of an existing deep learning framework, NucMM.
- Worked on developing a new joint-intensity and point-based, high throughput image registration algorithms having nanoscale precision (**10 – 40 nm**) for Multiplexed Expansion Revealing (**multiExR**). This work has been published at Nature Communications.

Invited Talks and Tutorials

- 03/26 **Comparative Analysis of Neural Population Codes**, *COSYNE 2026*
- 02/24 **Automating behavior quantification using deep learning**, *COSYNE 2024*
- 06/23 **SIS Symposium, Harvard SEAS**, *Seeing Beyond the Camera*

Honors & Awards

- 10/25 **NeurIPS 2025**: Student Travel (Scholar) Award
- 04/24 **NeuroMatch Academy**: Computational Neuroscience Summer School
- 02/24 **COSYNE 2024**: Undergraduate Travel Grant Award
- 04/23 **OxML 2023**: ML $\times$ Health Summer School, Oxford Mathematical Institute

Students Mentored

- **Pratham Yashwante**: Masters Student at UC San Diego
- **Chanyoung Park**: Undergraduate Student at UC San Diego

Technical Skills

Languages: Python, C, C++, MATLAB, Unix Shell Scripting, $\text{\LaTeX}$
Technologies/Frameworks: Keras, Tensorflow, Numpy, PyTorch, Git, OpenCV

References

Meenakshi Khosla: Assistant Professor of Cognitive Science and Computer Science and Engineering, UC San Diego. (Contact: mkhosla@ucsd.edu)

Edward S. Boyden: Y. Eva Tan Professor in Neurotechnology, MIT. (Contact: eboyden@mit.edu)

Satrajit S. Ghosh: Director of Open Data in Neuroscience Initiative and Senior Research Scientist, MIT. (Contact: satra@mit.edu)

Talmo D. Pereira: Salk Fellow and Principal Investigator, Salk Institute for Biological Studies. (Contact: talmo@salk.edu)

Donglai Wei: Assistant Professor of Computer Science, Boston College. (Contact: donglai.wei@bc.edu)